VI SEMESTER (JOB ORIENTED ELECTIVE – II)	L	T	P	C
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20CS6J02 - SOFTWARE TESTING TOOLS				

COURSE OUTCOMES:

At the end of the course students are able to

CO1: learn Manual testing techniques and software test levels
CO2: practice Java Programme for Selenium and Test frame works
CO3: learn Apache JMeter and Building a JMeter Test Plan
CO4: Running Multiple Scripts with JMeter and Different Types of JMeter Test Plans
CO5: Practicing JIRA and Test Management In JIRA (Using Zephyr Plug-in)

UNIT-I:

Manual Testing : Software Development Life Cycle (Requirements Gathering, Analysis and Planning, Software Design, Coding/Implementation, Testing, and Release and Maintenance Phase)
Software Test Levels (Unit Testing, Integration Testing, System Testing, and Acceptance Testing)

Software Test Types
Software Test Design Techniques
Software Test Life Cycle
Software Documents
Software Testing Standards
Software Testing certification/s

UNIT-II:

'Selenium with Java' : Java Programme for Selenium (Data Types, Variables, Operators, Control Flow, Strings, Arrays, IO, Methods, Exception Handling, and Object-Oriented Programming.)

- Selenium WebDriver (Web/HTML Elements, Inspecting Web Elements, Locating Elements, Selenium WebDriver API commands, Wait statements, and Page Object Model.
- TestNG Testing Framework (Create Test cases, Prioritise Test cases, Grouping Test Cases, Batch Testing, and Generating Test Results.)
- Automation Framework

UNIT-III:

JMeter:

- Introduction to Apache JMeter
- Elements of JMeter Test Plan
- Building a JMeter Test Plan
- Recording Tests Using JMeter
- Enhancements in Test Scripts

UNIT-IV:

JMeter Result Analysis

- Running Multiple Scripts with JMeter
- Different Types of JMeter Test Plans
- JMeter Distributed (Remote) Testing
- JMeter Functions, Variables and Regular Expressions
- JMeter Best Practices

UNIT-V:

Jira Tool Syllabus:

- Introduction of JIRA
- Getting started with JIRA
- Test Management In JIRA (Using Zephyr Plug-in)
- Defect Management In JIRA
- Advanced Search Using JQL
- Generating Reports In JIRA

Web References:

1. https://www.lambdatest.com/blog/selenium-with-java/
2. https://www.gcreddy.com/2021/09/apache-jmeter-syllabus.html
3. https://www.javatpoint.com/jira-tutorial